



A lightweight real-time karaoke system with improved singing voice separation via fundamental frequency estimation integration

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Abstract

With the fast-paced progress of deep learning, music source separation models have achieved impressive results. However, high separation quality often comes at the cost of complex architectures, limiting real-time performance. This study enhances singing voice separation by improving MMDenseNet and developing a real-time karaoke system that captures and separates accompaniment from played songs. We integrate traditional fundamental frequency (F0) estimation into complex ratio mask estimation, forming an integrated complex mask framework. On MUSDB18, we raise SDR from 3.87 dB to 6.44 dB over the original MMDenseNet (+2.57 dB absolute; +61% relative) while using only 0.33 M parameters—the smallest among compared methods. To enable real-time applications, we redesign the model with a shallow neural network that learns from traditional F0 estimation. The final system achieves real-time throughput (RTF < 1), e.g., 0.80 on an Intel i5-10500 PC and 0.88 on an NVIDIA Jetson AGX Xavier, demonstrating its potential for karaoke and related applications.

Keywords F0 estimation · Singing voice/accompaniment separation system · MMDenseNet · Integrated mask

1 Introduction

In the field of music information retrieval, Music Source Separation (MSS) is a primary research focus, with singing voice separation (also known as vocal and accompaniment separation) playing a crucial role. This technique has a significant impact on various downstream applications in music production, including lyrics recognition and alignment, kara-

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oke creation, and music remixing. The study of Music Source Separation (MSS) originated from Blind Source Separation (BSS), with early research primarily relying on traditional signal processing techniques. These methods often assume that a mixture spectrogram can be represented by a limited number of basis components and leverage algebraic approaches to decompose it into its original spectral templates and their temporal activations. Common techniques include Principal Component Analysis (PCA), Non-negative Matrix Factorization (NMF), etc. Finally, separation is typically achieved by deriving Wiener filters to estimate the target signal. While these traditional algorithms offer interpretability and achieve reasonable performance in controlled conditions, their effectiveness is constrained by several key limitations. First, they often rely on strong statistical assumptions, such as the independence of sources (ICA) or the non-negativity of spectral components (NMF), which may not always hold in complex music mixtures. Second, these methods struggle with highly overlapping frequency components, especially when the mixture contains a wide variety of accompanying instruments with similar spectral characteristics. Third, traditional approaches typically require prior knowledge or handcrafted constraints to guide separation, making them less adaptive to diverse musical styles and recording conditions. Due to these limitations, research has gradually shifted toward data-driven approaches, particularly deep learning-based methods, which can learn complex mappings directly from large-scale datasets. These methods, such as deep neural networks (DNNs), recurrent neural networks (RNNs), and more recently, convolutional and attention-based architectures, have demonstrated superior performance by leveraging large training corpora and powerful feature extraction capabilities. In the past few years,, several state-of-the-art models such as Demucs [1] and BSRNN [2] have been proposed, significantly improving the quality of music source separation, which has in turn boosted the performance of singing voice separation. Demucs is built on the U-Net [3] convolutional structure, drawing inspiration from Wave-U-Net [4]. Unlike traditional spectrogram-based approaches, Demucs operates directly on the waveform domain, leveraging a fully convolutional encoder-decoder architecture with skip connections to efficiently capture temporal dependencies. Additionally, it incorporates LSTMs between the encoder and decoder to enhance long-term modeling capabilities. Both Demucs and its subsequent versions have demonstrated impressive performance in MSS tasks, outperforming many spectrogram-based methods, especially in scenarios requiring high-quality audio reconstruction. Band-Split RNN (BSRNN) is a frequency-domain approach used in music source separation that utilizes a unique band-splitting and processing mechanism to improve the separation quality. BSRNN divides the audio input into multiple frequency bands, allowing each band to be processed independently through interleaved band-level and sequence-level modeling. This split-band approach enables the model to specialize in handling distinct frequency characteristics within each band, resulting in a more accurate and effective separation of sources like vocals and instruments. By focusing on specific frequency ranges, BSRNN achieves higher separation precision, significantly outperforming top-ranking models in the 2021 Music Demixing (MDX) Challenge [5]. However, despite the ability of these state-of-the-art models to isolate high-quality audio sources, their high parameter counts, latency, and CPU resource demands pose significant challenges for real-time processing in edge device applications. From this perspective, if the goal is real-time applications, a relatively lightweight model would be the ideal choice. MMDenseNet [6] is one such model; while its separation performance may not match state-of-the-art models, its

low parameter amount and high processing speed make it highly advantageous for practical applications.

Our main contributions in this paper are summarized as follows:

- We explore the integration of fundamental frequency (F0) estimation into singing voice separation, providing additional information to improve model robustness and accuracy.
- By incorporating F0 estimation, we refine the MMDenseNet architecture, leading to a more stable and effective separation model.
- We develop a real-time karaoke application optimized for both PC and NVIDIA Jetson AGX Xavier, achieving low-latency and high-performance separation suitable for practical deployment.

The organization of the paper is as follows: Sect. 2 introduces related works. Section 3 presents the proposed methods for improving MMDenseNet and adaption for the real-time application. Section 4 shows the evaluation metrics and experimental results. Section 5 gives a conclusion to this paper.

2 Related work

In this section, we review relevant research on fundamental frequency (F0) estimation and music source separation (MSS). We first discuss various F0 estimation approaches, followed by an introduction to MMDenseNet, a lightweight CNN-based model used as our baseline for MSS.

Sound consists of fundamental and harmonic frequencies, and singing is no exception. The fundamental frequency (F0) is a key element influencing a sound's tonal quality. Moreover, singing voices often carry the primary melodic content in music, making vocal F0 estimation an essential auxiliary feature [7]. This estimation is closely intertwined with singing voice separation, as the two tasks can reinforce and benefit from each other. Studies such as [8] and [9] have demonstrated that integrating F0 information from estimation algorithms can optimize separation performance. On the other hand, in [10], using the separated singing voice as input allows for more stable and accurate fundamental frequency estimation, thereby increasing the effectiveness and simplicity of the task. These studies have highlighted the beneficial relationship between these two tasks, showing how progress in one aspect can effectively support the other's performance. Prior F0-informed MSS methods typically use F0 to construct magnitude-only harmonic masks or introduce F0 as an auxiliary branch, applied on top of RPCA/NMF or U-Net-like separators. In contrast, our approach injects a full-band F0 salience map directly into an integrated complex ratio mask built on a lightweight MMDenseNet backbone, so that F0 jointly modulates both the magnitude and phase of the estimated spectrogram. This complex-domain integration of F0, together with the sub-million parameter budget, distinguishes our method from previous F0-informed systems and preserves real-time applicability.

Fundamental frequency (F0) estimation has been extensively studied, leading to the development of numerous algorithms ranging from traditional signal processing techniques to deep learning-based approaches. Common methods include YIN [11] and pYIN [12], which are time-domain techniques that detect periodicity in the signal and perform well for

monophonic inputs. Frequency-domain approaches, such as SWIPE [13], analyze harmonic structures but may struggle in noisy environments. Saliency-based methods, such as Melodia [14], operate in the time-frequency domain and are particularly effective in tracking prominent pitch contours. More recently, neural network-based models like CREPE [15] leverage deep learning to achieve high accuracy across diverse audio signals.

Given the importance of F0 estimation in music analysis and its potential to enhance MSS, we incorporate it into our approach to improve singing voice separation. MMDenseNet, which serves as our baseline separation model, adopts a multi-scale structure and effectively leverages CNNs to address the music source separation (MSS) problem. MMDenseNet is composed of multiple Multi-scale DenseNet modules, each of which contains several DenseBlocks, as illustrated in Fig. 1. These distinct Multi-scale DenseNet modules are designed to handle information from different frequency bands independently. The detailed architecture of MDenseNet is illustrated in Fig. 2, adopting an encoder-decoder structure composed of multiple DenseBlocks. Each DenseBlock consists of a series of network layers, including convolution, pooling, and batch normalization. Unlike the traditional ResNet structure, which applies simple addition to merge outputs, DenseBlocks concatenate all preceding layers' outputs. This design alleviates the vanishing gradient problem and enhances feature propagation. Additionally, skip connections between DenseBlocks are employed to capture long-context information from musical signals. After processing, the outputs from each module are combined to derive the spectrogram of the separated signal. While U-Net-based models rely heavily on large convolutional layers and transformer-based models like Hybrid Demucs [16] demand significant computational resources, MMDenseNet is designed with efficiency in mind. By leveraging the DenseNet architecture with its densely connected layers, the model promotes feature reuse, enabling information to flow directly from one layer to another without redundant computations. This design reduces the number of parameters while maintaining a strong representational capacity, leading to lower computational overhead. As a result, MMDenseNet strikes a balance between lightweight

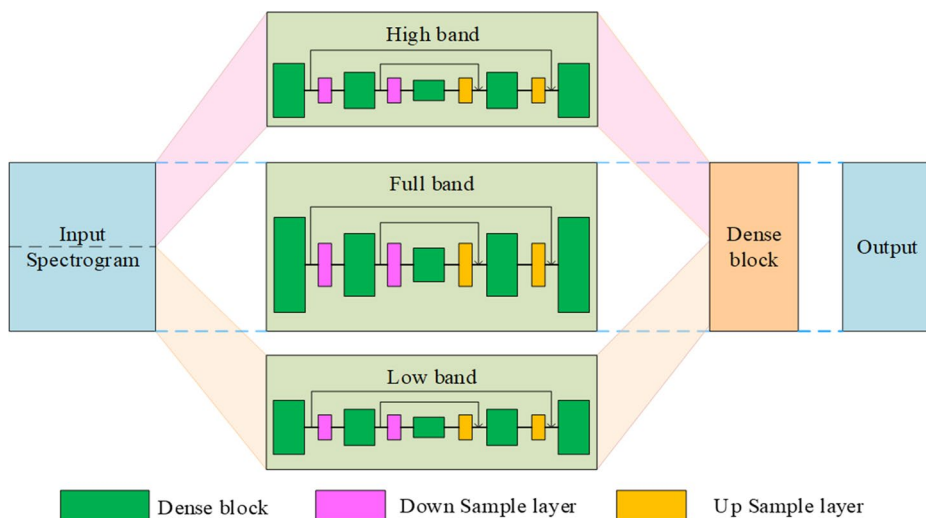


Fig. 1 Structure of the MMDenseNet

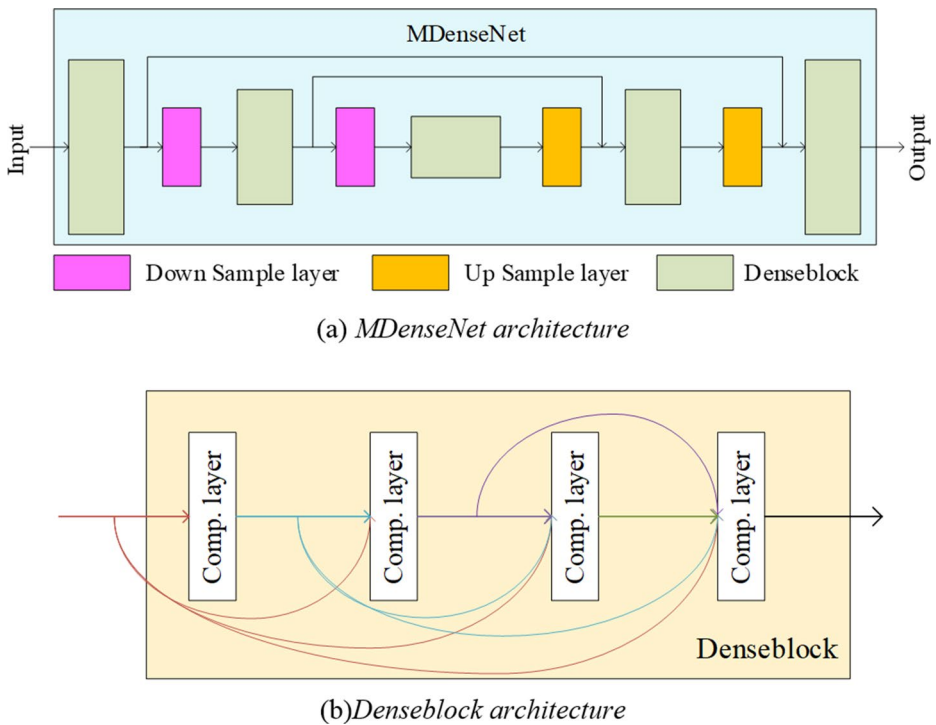


Fig. 2 (a) MDenseNet architecture (b) Denseblock architecture. The Comp. layer comprises a series of operations, including convolution, batch normalization (BN), and others

design and separation quality, making it particularly suitable for real-time applications on resource-limited devices.

3 Proposed system

This section will be divided into two parts. First, we introduce each network and module in the system, beginning with MMDenseNet. We demonstrate how its performance can be enhanced for improved separation and discuss the effects of these enhancements. Building on this foundation, we further illustrate the adaptations made to enable real-time processing for karaoke applications.

3.1 F0-integrated MMDenseNet

3.1.1 Efficient channel attention layer

In surround sound systems, vocals are often concentrated in the center channel, while other audio elements are distributed across the left and right channels. By focusing attention on the desired channel, vocal separation performance can be significantly improved. To achieve this, we leverage the ECA mechanism to capture channel-wise dependencies and

selectively emphasize channels that are more critical for vocal separation, thereby enhancing the model's ability to isolate vocal components effectively.

Efficient Channel Attention (ECA) [17] was first proposed in 2020. Unlike traditional channel attention mechanisms, it modifies the Squeeze-and-Excitation (SE) block [18]. In the SE block, global max pooling is applied to the height (H) and width (W) of the feature map, and the pooled vectors are processed through an MLP to generate channel-wise weights. ECA replaces the MLP in the SE block with a one-dimensional convolution, effectively reducing the number of parameters. ECA block considers the direct interaction between each channel and its k-nearest neighbors, i.e., how many neighbors participate in attention prediction of one channel.

$$F_{ECA}(x, \theta) = \sigma(\text{Conv1D}(\text{GAP}(x))), \quad (1)$$

where $\text{Conv1D}(\cdot)$ denotes 1D convolution with a kernel size k across the channel domain. The size of the kernel, k, is dynamically adjusted through:

$$k = \psi(C) = \left\lfloor \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma} \right\rfloor_{\text{odd}}, \quad (2)$$

where $|n|_{\text{odd}}$ represents the nearest odd number of n, and γ and b are respectively set to 2 and 1 in our work.

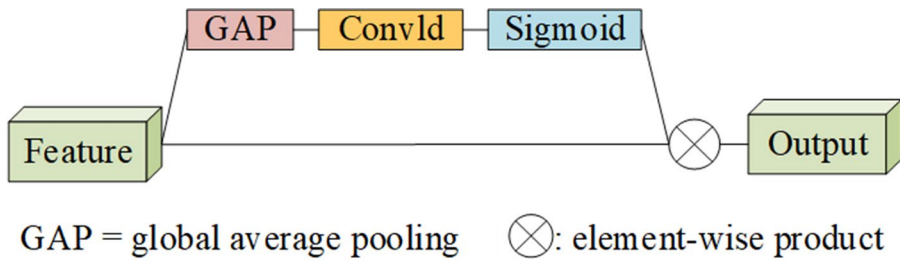
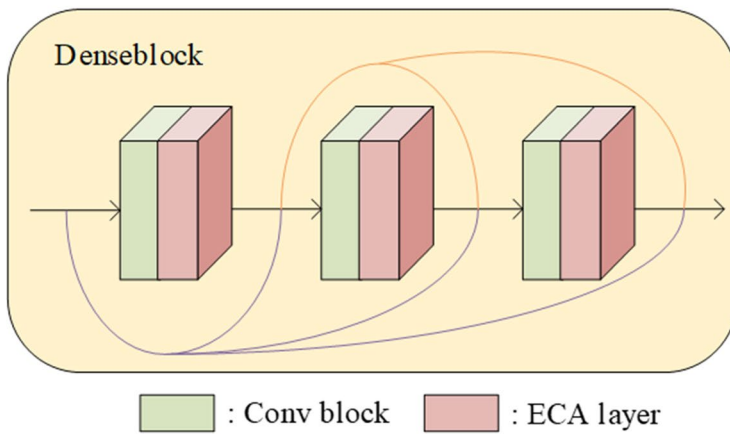
Regarding how the ECA layer helps improve model performance, we incorporate the ECA module after each convolution block within the encoder of every Denseblock. This approach enables the model to effectively capture channel-wise dependencies, as illustrated in Fig. 3.

3.1.2 F0-integrated complex mask

Given the discussion in Sect. 2, numerous studies have shown that F0 estimation and MSS tasks, particularly singing voice separation, can benefit each other. Various approaches have been proposed to integrate them. For example, [10] adopts a shared-weight joint model to simultaneously learn F0 estimation and MSS, while [8] incorporates F0 information directly into the estimated mask to enhance separation performance. In our work, we also adopt a mask-based approach, leveraging F0 information to refine the separation process.

There are various algorithms for fundamental frequency estimation, as mentioned in Sect. 2, including PYIN, SWIPE, and Melodia. A comparative study in [19] highlights the differences among these algorithms, showing that Melodia provides smoother outputs with fewer noise artifacts and discontinuities compared to the other two, as illustrated in Fig. 4. Additionally, [20] found that Melodia emphasizes vocal content effectively but often misses instrumental components. Since our goal is to separate vocals and accompaniment, this characteristic does not negatively impact our approach.

The Melodia algorithm consists of four main stages, as shown in Fig. 5: sinusoid extraction, salience function computation, pitch contour creation, and melody selection. Sinusoid extraction involves filtering and applying the Short-Time Fourier Transform (STFT). The filtering step is designed to enhance frequency ranges sensitive to human hearing, using a 10th-order IIR filter and a second-order Butterworth high-pass filter. After filtering, STFT is

(a) *Diagram of ECA module*(b) *ECA Denseblock architecture***Fig. 3** (a) Efficient Channel Attention Mechanisms and (b) Denseblock architecture with ECA layer placed after convolution block

applied, and the local maxima in the resulting spectrogram are identified as spectral peaks. Next, in order to avoid the error of the peaks being limited to the bin frequency of the FFT, the phase spectrum φ is used to estimate more accurate frequencies and values. The exact instantaneous frequency and amplitude are calculated as follows:

$$f_i = (\epsilon_i + v(\epsilon_i)) \frac{f_s}{N} \quad (3)$$

$$v(\epsilon_i) = \frac{N}{2\pi H} \Psi(\varphi_l(\epsilon_i) - \varphi_{l-1}(\epsilon_i) - \frac{2\pi H}{N} \epsilon_i). \quad (4)$$

The successive phase spectra will be used to estimate the peak instantaneous frequency in bin ϵ_i , while the bin offset will be calculated by the principal argument function Ψ . N and H denote the FFT length and hop size, respectively, which will be specified in the experi-

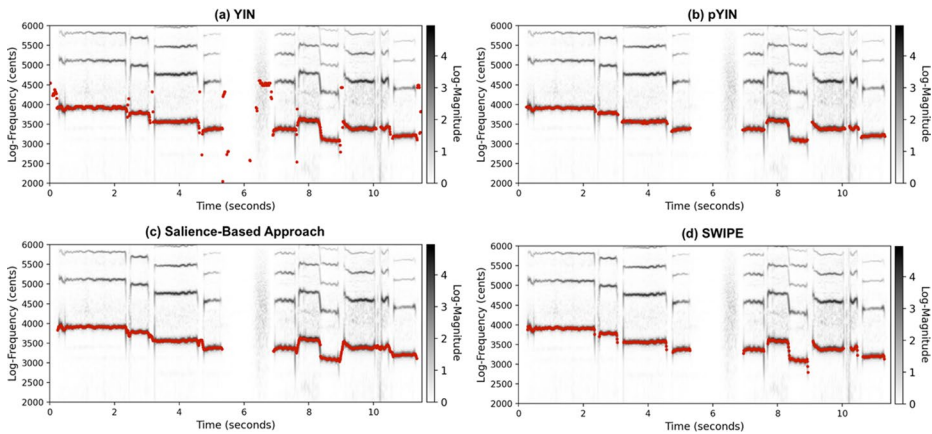


Fig. 4 Illustration of different F0-estimation algorithms. The log-frequency spectrograms (frequency values given in cents with reference frequency 55 Hz)

mental configuration. Further, the formula for instantaneous magnitude $\hat{a}_i$ is shown below, where W_{Hann} represents the Hann window and $X_l(\epsilon_i)$ denotes the spectral magnitude.

$$\hat{a}_i = \frac{1}{2} \frac{|X_l(\epsilon_i)|}{W_{Hann}\left(\frac{M}{N}v(\epsilon_i)\right)} \quad (5)$$

Saliency is determined using harmonic summation, where the saliency of a given frequency is calculated by summing the weighted energies present at its harmonic multiples, and the saliency function spans a pitch range from 55 Hz to 1.76 kHz, divided into 600 bins ($b=1 \dots 600$) with each bin representing 10 cents on a cent scale. The saliency function $S(b)$ is derived from frequencies f_i and linear magnitudes $\hat{a}_i$ obtained from the previous extraction step, where $i=1 \dots I$ and I represents the total amount of detected peaks. The bin $B(f)$ corresponding to frequency f and the formula for computing $S(b)$ are defined as follows:

$$B(f) = \lfloor 1200 \cdot \log_2\left(\frac{f}{55}\right) + 1 \rfloor, \quad (6)$$

$$S(b) = \sum_{h=1}^N \sum_{i=1}^I e(\hat{a}_i) \cdot g(b, h, f_i) \cdot (\hat{a}_i)^\beta, \quad (7)$$

where $e(\hat{a}_i)$ is a magnitude threshold function, β is compression parameter, and $g(b, h, f_i)$ denotes the weighting function. The magnitude threshold function is defined as follows:

$$e(\hat{a}_i) = \begin{cases} 1, & \text{if } 20 \log_{10}\left(\frac{\hat{a}_M}{\hat{a}_i}\right) < \gamma \\ 0, & \text{otherwise} \end{cases}. \quad (8)$$

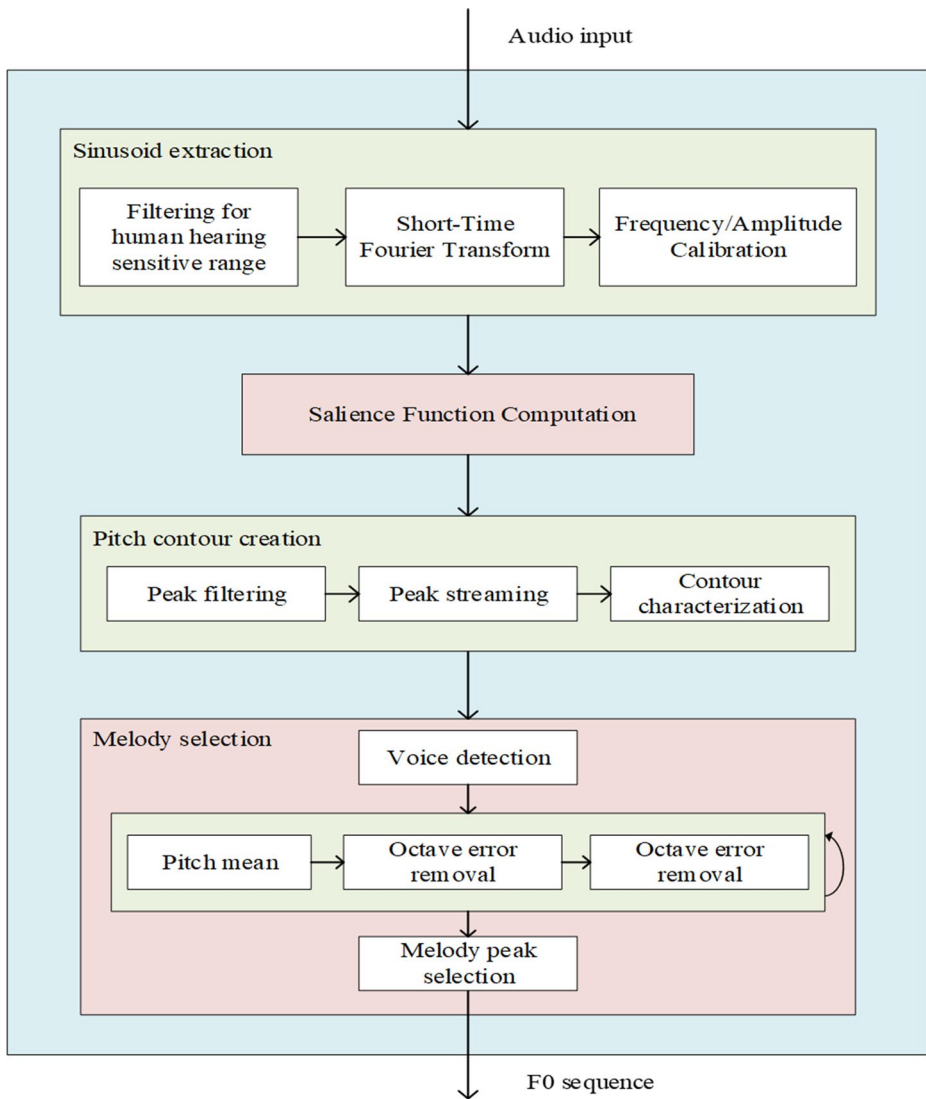


Fig. 5 Block diagrams of Melodia algorithm

The maximum difference between $\hat{a}_M$ and $\hat{a}_i$ is denoted by γ , where $\hat{a}_M$ indicates the magnitude of the highest spectral peak. The weighting function $g(b, h, f_i)$ is described below:

$$g(b, h, f_i) = \begin{cases} \cos^2(\delta \cdot \frac{\pi}{2}) \cdot \alpha^{h-1}, & \text{if } |\delta| \leq 1 \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

Here, h represents the harmonic order, while $\delta = |B(\frac{f_i}{h}) - b|/10$ stands for the semitone distance between the harmonic frequency and the center frequency of bin b , and α acts as the harmonic weighting factor.

After computing the salience function, the salient peaks at each time step are grouped into pitch contours, representing candidate fundamental frequencies, while non-salient peaks are filtered out to reduce noise. To ensure continuity, the frequency variation within each contour is restricted to a maximum of 80 cents. After generating the contours, the next step is to determine whether each contour belongs to the melody or not. This classification is based on several characteristics, including pitch mean and deviation, mean and total salience, contour length, and the presence of vibrato. In the end, F_0 is obtained through a three-step process: first, detecting whether the melody is present by analyzing pitch salience, contour characteristics, and vibrato patterns; second, refining the selection by identifying and removing octave errors and pitch outliers using melody pitch mean and trajectory continuity; and finally, selecting the most salient and continuous contour as the final melody representation.

After calculating the F_0 saliency information, the next step is to integrate it into the model's output. For this, we design the model to output a magnitude mask. However, in addition to estimating the magnitude, phase information is equally critical for music source separation (MSS), as demonstrated in [21]. Our integrated complex mask framework and system architecture are illustrated in Figs. 6 and 7, and the formula for the complex spectrogram signal estimation is as follows:

$$S = ReLU(|X| \odot \sigma(M) + D) e^{j(\angle \theta + \angle X)} \quad (10)$$

$$\cos \angle \theta = \frac{P_r}{\sqrt{P_r^2 + P_i^2}}, \sin \angle \theta = \frac{P_i}{\sqrt{P_r^2 + P_i^2}} \quad (11)$$

where M and D denote the magnitude mask and direct magnitude prediction, respectively. P_r and P_i are used for estimating the phase of the spectrogram, where P_r represents the real part and P_i represents the imaginary part. The phase θ is then calculated based on these two components.

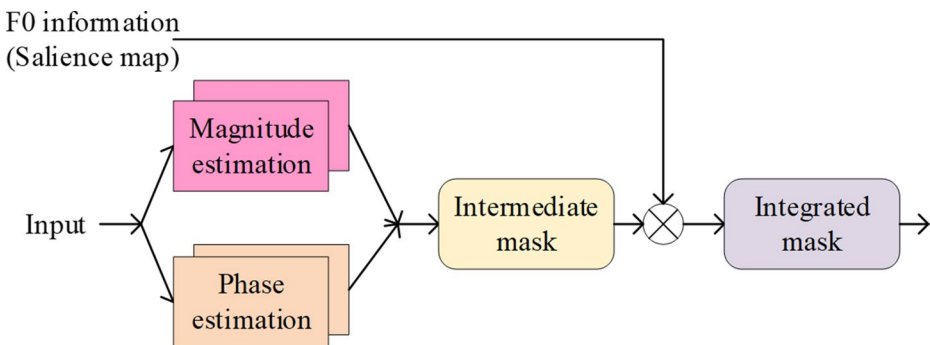


Fig. 6 Mask estimation structure

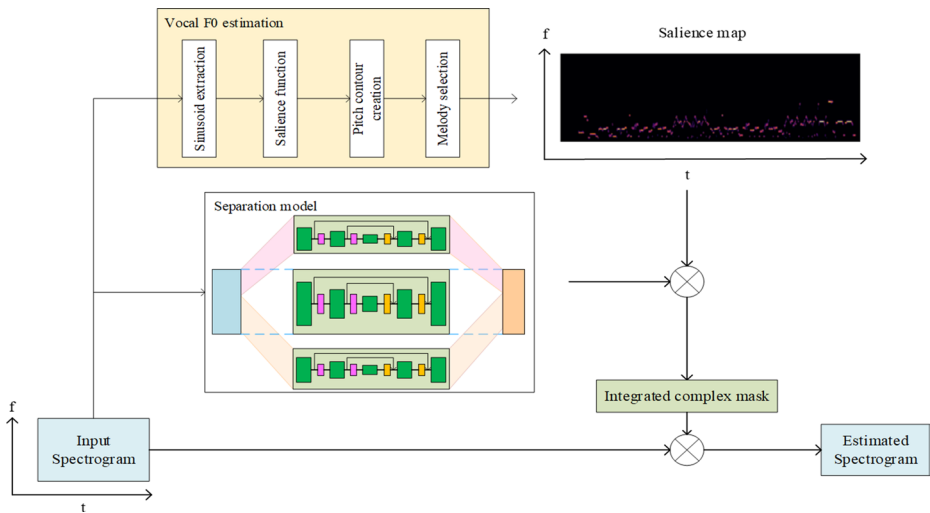


Fig. 7 Overview of model architecture

3.2 Real-time singing voice cancellation system

Music source separation (MSS) has a wide range of practical applications, one of which is karaoke generation. In this paper, we focus on the task of accompaniment separation to align with karaoke functionality. To enable real-time processing, we made targeted modifications to both the model architecture and training methodology.

3.2.1 Full band F0 estimation module

Due to the lengthy execution time of the Melodia algorithm, it is unsuitable for real-time processing. To address this, we designed a lightweight full-band MDenseNet to approximate the outputs of the Melodia algorithm. The model structure for predicting F0 is shown in Fig. 8. This approach is conceptually similar to knowledge distillation, where the Melodia-generated saliency maps serve as weak ground truth. While these maps are not perfect representations of the true ground truth, they effectively guide the model in extracting F0 information. During training, the F0 model's loss is jointly optimized along with the primary separation model, ensuring that F0 prediction and separation performance are improved simultaneously.

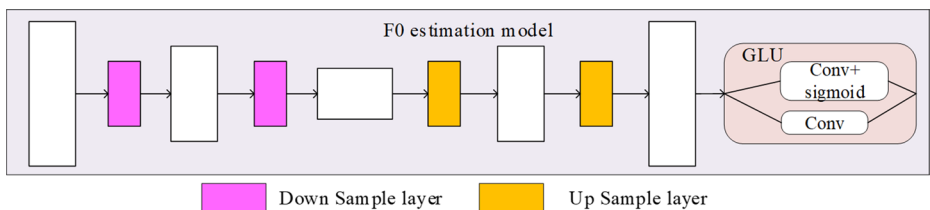


Fig. 8 Architecture of F0 estimation module

3.2.2 Real-time processing procedures and training adjustment

The importance of maintaining an appropriate buffer size, as emphasized in [22], is crucial for reducing CPU load and enabling block processing. Based on [22, 23], we implemented a operation flow of streaming processing, as depicted in Fig. 9. To accommodate the shorter pending buffer length required for real-time processing, data preprocessing and adjustments to the training length were necessary. To address this, we employed a source activity detector to identify the main segments of the audio data. Specifically, the audio was divided into smaller training chunks, and the RMS (Root Mean Square) value of each segment was calculated. Segments with RMS values exceeding 15th percentile of the RMS values were selected as the primary active fragments, reducing the training data length.

4 Experimental setup & results

This section provides a comprehensive description of the implementation details. We cover the chosen hyperparameters, the preparation and usage of training dataset, the evaluation metrics employed, and results analysis.

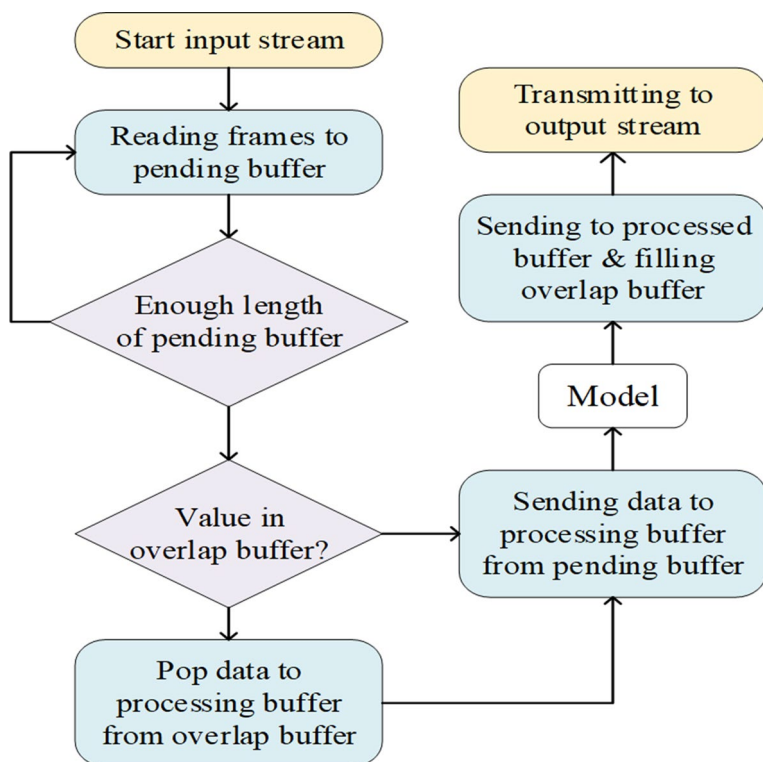


Fig. 9 Real-time processing procedures

4.1 Dataset

We use the MusDB18 [24] dataset, which consists of 150 stereo songs spanning various genres, sampled at 44,100 Hz, along with their isolated drums, bass, vocals, and other stems. MusDB18 is divided into 86 tracks for training, 14 tracks for validation, and 50 tracks for the test set. Supervised methods are typically trained on the training set, while performance is evaluated on the validation and test sets.

4.2 Experimental setup

The training process is conducted over 200 epochs with a batch size of 8. We employ the Adam optimizer [25] to update the model parameters throughout the training period. Additionally, random channel swapping and random gain are applied as data augmentation techniques to increase the robustness of the model, while no F0-specific data augmentation (e.g., pitch shifting or synthetic vocal generation) is employed in our experiments. The initial learning rate is set to 0.001 and is adjusted dynamically using a scheduler. This scheduler tracks validation loss and adjusts the learning rate by multiplying it with 0.95 if there is no improvement over 2 consecutive epochs. The learning rate reduction continues until it reaches the minimum threshold of 1×10^{-5} . We perform a short-time Fourier transform (STFT) on the training data using a Hann window with a size of 2048 and a hop size of 1024. The training loss is calculated using Mean Absolute Error (MAE) loss. If the F0 Estimation module is included, the F0 prediction loss is incorporated into the total loss.

4.3 Evaluation metrics

Performance measurements for source separation models were proposed by Vincent et al. [26] for blind source separation and later adapted for many supervised source separation campaigns. Most studies use signal-to-distortion ratio (SDR) to calculate the difference between the predicted result ($\hat{s}_j$) and the ground truth signal (s_j) in two steps. Starting from the decomposition of $\hat{s}_j$ as:

$$\hat{s}_j = s_j + e_{interf} + e_{noise} + e_{artif}, \quad (12)$$

where e_{interf} , e_{noise} and e_{artif} represent the interference, noise, and artifact error terms, respectively. Based on this definition, the source-to-distortion ratio (SDR) value is then computed.

$$SDR = 10 \log_{10} \frac{\|s_j\|^2}{\|e_{interf} + e_{noise} + e_{artif}\|^2} \quad (13)$$

To further analyze the error components, we also report the signal-to-interference ratio (SIR) and the signal-to-artifacts ratio (SAR). SIR evaluates how well the method suppresses interference from the other sources, whereas SAR measures the number of artifacts introduced by the separation algorithm. They are defined as

$$SIR = 10 \log_{10} \frac{\|s_j\|^2}{\|e_{interf}\|^2} \quad (14)$$

$$SAR = 10 \log_{10} \frac{\|s_j + e_{interf}\|^2}{\|e_{artif}\|^2} \quad (15)$$

Together with SDR, these three metrics provide a comprehensive evaluation of both the overall separation quality and the nature of the remaining errors.

4.4 Ablation studies

In this subsection, we perform a series of ablation experiments to quantify the contribution of each module to the overall system performance. We first evaluate the impact of the F0 module and the efficient channel attention (ECA) blocks on the MUSDB18 dataset by measuring the signal-to-distortion ratio (SDR) under different combinations of these modules. As shown in Table 1, we compare the SDR obtained when only the F0 module is enabled, when only ECA is used, and when both components are jointly integrated, for both vocal and accompaniment separation. We then focus on the proposed integrated complex mask and analyze several variants of its internal submodules; the corresponding results are summarized in Table 2. The results in Table 2 show that removing either the phase term or the residual branch D consistently degrades SDR, confirming that the full complex mask design is beneficial.

Table 1 Ablation of ECA and F0 integration on MUSDB18 vocal and accompaniment separation. All values are median SDR (dB). “ECA” and “F0” indicate whether the efficient channel attention and the F0-integrated complex mask are enabled

Model	F0	ECA	SDR_voc(dB)	SDR_acc(dB)
Complex MMDenseNet (ECA + F0)	✓	✓	6.44	13.38
Complex MMDenseNet (only F0)	✓	✗	5.86	12.41
Complex MMDenseNet (only ECA)	✗	✓	4.76	10.01

Table 2 Ablation of the proposed complex mask variants on the MUSDB18 dataset. All values are median SDR (dB) on the vocal stem

Type	SDR(dB)
Complex Mask $(ReLU(X \odot \sigma(M) + D) e^{j(\angle\theta + \angle X)})$	6.43
Complex Mask $(ReLU(X \odot \sigma(M) + D))$	5.46
Complex Mask $(ReLU(X \odot \sigma(M)) e^{j(\angle\theta + \angle X)})$	5.78
Complex Mask $(ReLU(D) e^{j(\angle\theta + \angle X)})$	5.86

4.5 Comparison with other methods

We next compare the proposed method with other music source separation (MSS) models under the same MUSDB18 test-set protocol. In all cases, the reported SDR values correspond to the 50 test songs of the MUSDB18 dataset. For our system, SDR (and, when available, SIR and SAR) are computed using the *museval* implementation of the BSS Eval metrics, whereas the scores of the comparison methods are taken from their original publications, which also rely on BSS Eval or equivalent configurations.

Table 3 summarizes the median vocal SDR and the number of trainable parameters for all compared systems. Compared to the original MMDenseNet, the proposed model significantly improves vocal separation performance, with a relative gain of approximately 61%. Although our vocal SDR is not the highest among all methods, our model uses by far the smallest number of parameters, making it attractive for real-time deployment.

Table 4 focuses on the accompaniment stem and compares the accompaniment SDR of different MSS systems in the same two-stem (vocal + accompaniment) setting. When the modified full-band F_0 estimation module is integrated into our real-time karaoke model, the accompaniment SDR decreases only slightly (from 13.59 dB to 13.38 dB). This mild degradation is a direct consequence of the F_0 module emphasizing pitch-related harmonic structures in the vocal estimate, which shifts more energy from overlapping harmonic regions of the accompaniment into the vocal stem. Nevertheless, the accompaniment SDR remains high and competitive with other methods, while the gain on the more challenging vocal stem is substantial. We regard this trade-off as reasonable for karaoke-oriented usage, where residual singing in the backing track is generally more disturbing than small distortions in the accompaniment.

Table 3 Comparison with other methods. Vocal separation performance and model size of the proposed method and existing MSS systems on the MUSDB18 test set. All values are median SDR (dB) on the vocal stem. “Parameters (M)” denotes the number of trainable parameters in millions

Architecture	Vocal SDR(dB)	Model parameters(M)
MMDenseNet [27]	3.87	0.33
Wave-u-net [28]	3.25	10.2
Wavenet [29]	3.46	6.3
Jukebox [30]	5.06	N/A
Conv-TasNet [31]	6.43	9
DemucsV2 [31]	6.84	288
Open-Unmix [32]	6.32	35.55
Stripe-T [33]	7.75	10.06
BSRNN [2]	10.01	26
Ours	6.44	0.33

Table 4 Comparison with other methods. Accompaniment separation performance of the proposed method and existing MSS systems on the MUSDB18 test set. All values are median SDR (dB) on the Accompaniment stem

Architecture	SDR(dB)
Architecture	SDR(dB)
MMDenseNet [27]	15.41
Wave-u-net [28]	11.87
Wavenet [29]	10.64
Open-Unmix [32]	11.66
Ours (w/o modified F_0 module)	13.59
Ours (w/ modified F_0 module)	13.38

To provide a more detailed error analysis in this two-stem configuration, we also report SIR and SAR for our model. Using the *museval* implementation of BSS Eval, the proposed vocal separation system achieves a median SIR of 17.11 dB and a median SAR of 6.34 dB on the MUSDB18 test set. These results indicate that the model can strongly suppress interference from the accompaniment while introducing only a moderate level of separation artifacts. Note that many prior works report SIR and SAR in a four-stem configuration; therefore, these values are not directly comparable to those multi-stem results and are mainly intended to characterize the behavior of our system beyond SDR.

4.6 Comparison with other real-time karaoke system

In addition to comparing our model with other MSS architectures, we also consider a system-oriented real-time karaoke system. Tachibana et al. [34] propose a real-time audio-to-audio karaoke generation system for monaural recordings, which combines a two-stage harmonic/percussive sound separation (HPSS) procedure for singing-voice suppression with a spectrogram-stretching, phase-vocoder-based pitch-shifting pipeline for key conversion. Their system is evaluated on the MIR-1 K dataset [35], a 1,000-clip karaoke corpus originally recorded with accompaniment and singing voice on separate channels; the clips are extracted from 110 karaoke songs and provide paired mixture and clean accompaniment tracks. In [34], accompaniment quality is assessed using an SDR measure consistent with our definitions in Eqs. (16)–(15). To provide a fair system-level comparison, we therefore evaluate our method on MIR-1 K under the same configuration and SDR metric. As summarized in Table 4, our Fo-informed MMDenseNet achieves clearly higher accompaniment SDR than the HPSS-based karaoke system of [34], indicating substantially improved backing-track quality while preserving real-time operation.

$$SDR(E(t), X(t)) = 10 \log_{10} \frac{\|aX(t)\|^2}{\|E(t) - aX(t)\|^2} \quad (16)$$

$$= 10 \log_{10} \frac{\langle X(t), E(t) \rangle^2}{\|X(t)\|^2 \|E(t)\|^2 - \langle X(t), E(t) \rangle^2}. \quad (17)$$

$E(t)$ and $X(t)$ represent the estimated signal and the ground truth, respectively. The notation $\langle \cdot, \cdot \rangle$ denotes the inner product, and the norm is defined as $\|x\|^2 = \langle x, x \rangle$. We adopt this SDR definition for our MIR-1 K experiments to ensure a fair comparison with the real-time karaoke system in [34].

4.7 Real-time evaluation

We computed the Real-Time Factor (RTF), defined as the processing time divided by the signal duration, on an Intel i5-10500 CPU and an NVIDIA Jetson AGX Xavier under a streaming setting to better reflect real-world conditions. During the experiment, we directly captured songs playing on YouTube and performed real-time separation to output the vocal-removed accompaniment. To ensure smooth streaming processing, we set the block size to 4000 samples and configured the pending buffer length to 10 blocks to minimize initial

latency. Additionally, the processing and processed buffers were set to 20 blocks each to maintain optimal separation performance. The RTF is 0.80 on the Intel i5-10500 CPU and 0.88 on the NVIDIA Jetson AGX Xavier.

4.8 Brief discussion on performance-efficiency trade-off

Our model strikes a strong balance between SDR and parameter efficiency—outperforming lightweight baselines and approaching larger systems such as DemucsV2—yet a gap to recent top performers (e.g., BSRNN, 10.01 dB) remains. We investigated stronger modules by inserting a self-attention block into MMDenseNet (pre-LayerNorm before attention) to capture long-range temporal context; however, under streaming constraints this increased parameters by $\approx 3\times$ while separation quality dropped to roughly half of the CNN baseline and latency budgets tightened. Given these observations, we retain a CNN-centric design for real-time stability.

5 Conclusions

In this paper, we proposed the integration of fundamental frequency estimation into MMDenseNet and developed a real-time karaoke system deployable on consumer-grade CPUs and edge devices like the AGX Xavier. This system allows users to obtain separated accompaniment in real-time with low latency. Comparative evaluations with other MSS methods demonstrated that our approach achieves competitive performance with the smallest parameter count, showcasing its effectiveness in improving separation quality. For future work, we aim to further enhance the model's performance while reducing computational complexity and latency, and to complement the real-time factor analysis with perceptual evaluations, such as subjective listening tests (e.g., MOS or MUSHRA) and objective perceptual metrics (e.g., PESQ, POLQA, STOI, or PEASS-style scores), in order to more directly assess the user experience of the proposed karaoke system.

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Data availability The datasets employed and examined in this study are available online on <https://zenodo.org/records/1117372> and <https://zenodo.org/records/3532216>.

Declarations

Ethics approval This is not relevant because this study didn't use any human or animal participants.

Consent to participate This is not relevant because no human or animal participants were used.

Consent for publication Publication didn't require consent because there were no human participants.

Competing interests The authors declare that they have no conflict of interest.

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